Deva Anand

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Education

University of Massachusetts Amherst — **September 2025 – May 2027 (Expected)**
Master of Science in Computer Science — Amherst, MA
Coursework: Advanced Machine Learning, Optimization in Computer Science, Advanced NLP, Research Methods in CS, Systems for Data Science

Vels Institute of Science, Technology and Advanced Studies — **August 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering — Chennai, India

Publication

- *"Alzheimer's Disease Classification using Transfer Learning,"* **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning to neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
ML & Deep Learning	PyTorch, JAX, TensorFlow, HuggingFace Transformers, NumPy, scikit-learn, reverse-mode autodiff, SGD/Adam optimization, regularization, normalizing flows, diffusion, transfer learning
Statistics & Experimentation	probability & statistics, hypothesis testing (bootstrap CIs, McNemar's), convergence & asymptotic analysis, experimental design, ablations, evaluation methodology, holdout protection
LLM / NLP / Retrieval	LLM & transformer evaluation, mechanistic interpretability, RAG, RRF, cross-encoder reranking, MMR, Ollama, llama-index, spaCy
Tools & Cloud	Claude / Claude Code, Cursor, GitHub Copilot, Weights & Biases, Git, GitHub Actions, Docker, AWS, GCP, Azure

Research & Projects

Generative Models & Deep Learning Research — github.com/Deva-1903/UMASS_CICS_689
JAX, PyTorch, NumPy | UMass CS 689 Advanced Machine Learning

- Trained and benchmarked **5 deep architectures** (perceptron, deep MLP, ReLU MLP, VGG-style CNN, ResNet) on CIFAR-10 across **3 optimizers** with learning-rate tuning, diagnosing convergence, overfitting, and generalization from train/test-error curves.
- Implemented generative models from scratch in JAX, a **normalizing flow with coupling layers** and a **DDPM diffusion model** (deriving the ELBO and Gaussian-KL objectives), and built a from-scratch **reverse-mode autodiff engine** in NumPy validated against JAX to machine precision.

Refusal Decay - Mechanistic Interpretability of LLM Safety — github.com/Deva-1903/refusal-decay
PyTorch, HuggingFace Transformers, Llama-3.1-8B | UMass COMPSCI 602 Research Methods (full marks)

- Investigated how prefilling attacks weaken refusal behavior in **Llama-3.1-8B-Instruct**, extracting the residual-stream refusal direction via difference-in-means on a held-out prompt set and tracing per-layer projections to understand model behavior under attack.
- Designed causal interventions (activation patching, additive direction injection via forward hooks) with random/orthogonal and benign controls; reported behavioral refusal dropping **0.92 to 0.32** and a clean null (0/300 recovered) using **bootstrap 95% CIs and McNemar's test**, with labels validated by a secondary classifier (**87.5% agreement**).

AutoEval - Retrieval Evaluation Improvement Agent — github.com/Deva-1903/AutoEval
Python, IR Evaluation, Git Automation

- Built a reusable evaluation-improvement agent that proposes new queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve **discriminative power** on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates to Git with report cards for an auditable, reproducible evaluation workflow.

KG2RAG-Enhanced - Multi-Hop QA on HotpotQA — github.com/Deva-1903/KG2RAG-685-NLP
Python, Ollama, sentence-transformers, llama-index, PyTorch | UMass CS 685 Advanced NLP

- Extended a knowledge-graph RAG pipeline with **multi-view seed retrieval**: generated single-hop sub-questions, retrieved passages per view, and fused results via Reciprocal Rank Fusion (RRF), cross-encoder reranking, and Maximal Marginal Relevance (MMR).
- Replaced heuristic top-M selection with a **0–1 knapsack** token-budget formulation (value = relevance $\times$ coverage); ran **100–500-question** batch experiments with token and latency logging on an Ollama + LLaMA-3 inference pipeline.

Experience

Wysa — **July 2023 – July 2025**
Backend Engineer — Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models, replacing spreadsheet workflows and cutting manual annotation by **100+ hours/month** while improving labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability with clean, modular services, integration tests, and **20+ VAPT** remediations.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023** (OSS contributions to Linkfree, Freehit, ProjectsHut); tutored **30+** students in programming at Sayur, a non-profit in South Tamil Nadu.